Akarsh Reddy Tatimakula

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Education

University of Colorado Boulder
Master of Science, Data Science *GPA: 3.97/4.0*
- Coursework: Machine Learning, Neural Networks, Information Visualization, Datacenter-Scale Computing, Data Mining

Vellore Institute of Technology (VIT)
Bachelor of Technology, Computer Science and Engineering *CGPA: 8.37/10*
- Coursework: Artificial Intelligence, Natural Language Processing, Cloud Computing, Data Analytics, Data Structures and Algorithms

Experience

Associate AI Engineer – FocalCXM LLC June 2025 – Present
- Architecting and deploying scalable AI agents using **Crew AI**, **LangChain**, **LangGraph**, and **AWS SageMaker**, orchestrated via optimized **RAG** and **MCP** workflows on **AWS (Lambda,Sagemaker)**, reducing manual processing time by 50%.
- Integrating **GPT-4o** and custom embeddings with RESTful APIs, **Amazon DynamoDB**, and **Pinecone** vector DB for high-throughput automation, improving intent classification accuracy by 35%.
- Fine-tuning LLMs and embeddings with iterative **Prompt engineering** and **Model Evaluation and Optimization** for product lead generation.

Machine Learning Researcher (Capstone Project) – Honda January 2025 – June 2025
- Designed and implemented an **LLM** analytics platform leveraging **OpenAI**, **DeepSeek**, and **LLaMA** with advanced prompt engineering and adaptive feedback loops, reducing report generation time by 50%.
- Engineered transformer benchmarking pipelines targeting latency and interpretability metrics, optimizing model configurations for 30% faster inference throughput.Collaborated with domain experts to validate and refine outputs, driving a 40% increase in enterprise report adoption.

Data Scientist – Serentica Renewables (KKR and Co) May 2023 – August 2023
- Forecasted market trends from **Indian Energy Exchange (IEX)**, increasing prediction accuracy by 25%.
- Refactored Python codebase with **NumPy** vectorization, reducing processing time by 20% and improving pipeline efficiency.
- Integrated time-series insights into live dashboards, streamlining decision-making for energy trading operations with **stakeholders**.

Data Scientist – Corizo September 2022 – November 2022
- Engineered customer segmentation and prediction pipelines using **TensorFlow**, **PyTorch**, **XGBoost**, **Hugging Face Transformers**, **GANs**, and **Graph Neural Networks**, boosting predictive performance by 18%.
- Automated **ETL** workflows **Apache Airflow**, **Kubeflow Pipelines**, and **MLflow**, cutting data preparation time by 40% and ensuring end-to-end reproducibility.
- Created interactive dashboards with **Streamlit** and **Plotly** to visualize insights, informing client strategy and increasing project adoption by 30%.

AI/ML Engineer – NeoGeoInfo Technologies August 2019 – August 2022
- Developed production-ready chatbots using **Rasa** and **Dialogflow**, defining intents/entities, agent-handoff rules, and backend integrations via **FastAPI**/**Flask**. Built **Generative AI** and NLP pipelines using **scikit-learn**, **spaCy**, and Transformers (**BERT**/**DistilBERT**).
- Implemented precision/recall/F1 evaluation with **MLflow**, containerized services using **Docker**, automated training pipelines with **Airflow**, deployed on **AWS (EC2, Lambda, SageMaker AI)**, and developed **Tableau** dashboards for key KPIs.

Skills

Programming & Data Analytics: Python, R, SQL, ETL Pipelines, Pandas, Web Scraping, Excel (Pivot Tables, VLOOKUP, VBA), Data Analysis, Statistical Analysis, Predictive Modeling, Data Visualization, Tableau, Power BI
Machine Learning & AI Frameworks: Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-Learn, Generative AI (GenAI), Large Language Models (LLMs), Agentic AI, CrewAI, LangChain, LangGraph, Fine-Tuning, Prompt Engineering, A/B Testing
Cloud & Data Platforms: AWS (S3, Lambda, SageMaker AI, Bedrock), Docker, API Integration, DynamoDB, Databricks, Pinecone Vector DB

Projects

AI Voice Assistant for Go-To-Market Automation (LiveKit, OpenAI, Whisper)
- Built a AI voice assistant using **LiveKit**, **OpenAI LLMs**, and **Whisper** for streaming **STT/TTS**, enabling automated product inquiries, natural interactions, and direct **lead capture** across **GTM** workflows/actions and triggering automated **CRM** updates
- Developed multi-turn dialogue pipelines with **intent extraction** and **next-best-action** logic, converting conversations into structured conversation.

Lead Generation AI Agent
- Delivered a **LangChain**-based AI agent solution using **Amazon S3**, **Pinecone** vector DB, **AWS SageMaker** for model hosting, and **AWS Lambda** for orchestration, automating lead extraction and increasing pipeline volume by 50% having a Follow-up Agent using **AWS(SES/SQS)**.
- Built **RAG**-powered parsing modules to convert unstructured emails and web data into qualified leads, improving lead accuracy by 40%.

Healthcare Insurance MCP Chatbot (Advanced RAG + Multi-Agent, AWS)
- Built a **MCP** chatbot with function calling and 95% confidence; returns schema-validated JSON (**Pydantic**) for Healthcare Insurance Claims.
- Implemented **RAG** on AWS: **Amazon Bedrock Titan** embeddings + **FAISS VertorDB** grounding **Anthropic Claude 3 Sonnet** for accurate, auditable answers with reduced **hallucination** rate.
- Shipped on **AWS (Lambda**,**SQS**,**SageMaker)** with a **Streamlit** UI; added eval harness (dataset replays) and prompt version control.

Research Publication

Evaluating the Role of AI and Big Data in Consumer Engagement, Australian Business Deans Council (ABDC) 2025

Spell Checker: Automated Spell-Checking Algorithms, Crimson Publishers 2022